Mitigating Preference Conflict in Controlled Decoding: A Principal-Agent Perspective

Haichuan Wang, Henry Huang, Emira Ibrahimović, Wilson Cheung

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Abstract

Misalignment at test time is fundamentally an incentive problem: the model's intrinsic preferences over outputs need not coincide with the user's true utility function. Existing test-time alignment methods overlook this incentive structure and presume that users will truthfully report their reward function. However, truthful revelation generally fails to maximize user utility, theoretically yielding suboptimal alignment, and we show that strategic manipulation is both feasible and practically implementable, making it a realistic concern. Moreover, naively maximizing user utility is computationally intractable, and imperfect attempts to approximate it can induce severe reward hacking, leading to undesirable behaviors and safety risks.

In this work, we introduce a test-time alignment framework grounded in principal–agent theory from algorithmic contract design and controlled decoding. We model the entity who provides the reward function as the principal and treat the language model as the agent, and we characterize the optimal reward function that maximizes the principal's utility while remaining robust to reward hacking. Our method satisfies several desired theoretical properties—including incentive compatibility and bounded KL deviation from the baseline policy—and we empirically demonstrate its advantages over existing test-time alignment approaches. Our formulation may also be of independent interest, as it extends classical principal–agent models to settings in which the agent operates under a KL-based entropy constraint.

1 Introduction

Large language model trained on huge amount of data has demonstrated impressive capabilities across a wide range of tasks. However, they still produce harmful and hallucinated content, preventing its adoption in real world adoptions. Therefore, it is critical to ensure that the output of the large language model is aligned to human values.

Alignment methods can be broadly categorized into generator improvement and inference-time alignment. Generator improvement uses reward models trained on human preference data to fine-tune language models [21, 26], typically via reinforcement learning methods such as PPO, though such training is often costly and unstable [24, 25]. In contrast, inference-time alignment modifies model behavior at generation time. A prominent approach is controlled decoding [19], which adjusts the baseline model's logits using the reward signal to guide token selection.

In this paper, we interpret controlled decoding through the lens of the principal–agent model in algorithmic contract design. In this framework, a principal seeks to induce an agent to take actions that may not align with the agent's own preferences; to do so, the principal designs an incentive scheme—typically formalized as a contract—that aligns the agent's behavior with the principal's objectives. Under this perspective, misalignment becomes an incentive-design problem: the model may favor responses that humans do not prefer, and the reward function serves as an implicit contract that encourages desirable outputs while discouraging undesirable ones. This viewpoint allows us to leverage tools from principal–agent theory to derive reward functions that optimally align the model to the principal's preference.

The core contribution of our approach is to connect algorithmic contract design with inference-time alignment. We derive the optimal reward function and apply reward shaping prior to using the reward signal in controlled decoding. Unlike prior work on controlled decoding, our method operates along an "orthogonal dimension": we reshape the reward and train the value function on this modified signal. As a result, our approach introduces no additional inference-time overhead and can be seamlessly integrated as a modular add-on to existing controlled decoding techniques.

To summarize our work: (1) We show that truthfully submitting one's reward function does not, in general, maximize the principal's utility, implying that truthful reporting incurs a utility loss. (2) To address this, we formulate test-time alignment through the principal–agent lens and derive an optimal reward function. Our approach can be incorporated into existing controlled decoding methods without introducing inference-time overhead. (3) We prove that the proposed method satisfies key theoretical properties, including incentive compatibility and bounded KL deviation from the baseline policy, and we demonstrate strong empirical performance relative to existing baselines. (4) We discuss the safety implications and applications of our framework and inference time alignment algorithm more broadly.

2 Related Works

Inference Time Alignment Recent work has explored aligning large language models at inference time, offering an alternative to computationally expensive fine-tuning approaches. These methods adjust model behavior during generation without modifying the underlying parameters, providing flexibility to adapt to different alignment objectives. One line of work [15] proposes a representation editing approach that views autoregressive LLMs as discrete-time stochastic dynamical systems, introducing external control signals into the model's state space. By training a value function on hidden states according to the Bellman equation, this method optimizes control signals at test time through gradient-based optimization, achieving competitive performance with fine-tuning methods while requiring significantly fewer computational resources. Another approach [20] frames the inference-time alignment problem as a variant of the multi-armed bandit problem, where a decision maker sequentially selects tokens to maximize cumulative rewards. Both approaches highlight the potential of achieving alignment at inference time without requiring model fine-tuning.

Principal Agent Model and Algorithmic Contract Design The principal–agent problem examines how a principal designs incentive schemes—typically referred to as contracts—to align the agent's interests with their own. Principal–agent problems have been extensively studied in economics [12, 9, 8], leading to conceptual breakthroughs that is recognized with 2016 Nobel Prize in Economics. In recent years, their algorithmic aspects have attracted significant attention from the computer science community, leading to a growing theory and ML literature on their computational properties [6, 23] and inspiring several approaches that apply contract-based ideas to align agents in multi-agent settings [11, 13].

Two lines of work are most closely related to ours. The first is Hadfield-Menell and Hadfield [10], which interprets the reward model in alignment as a contract and explains reward hacking through the lens of incomplete contract. However, Hadfield-Menell and Hadfield [10]'s argument remains largely conceptual, whereas we formulate a principal agent optimization problem and provide an implementable framework for improving test-time alignment. The second is Ben-Porat et al. [3], which theoretically studies how a principal can subsidize rewards on selected states in an MDP to induce desired behavior from the agent. Our formulation differs in that the agent's best response is subject to an entropy regularization, yielding different optimal solutions, and our work is grounded in alignment application.

3 Problem Formulation: Controlled Decoding from a principal-agent perspective

3.1 Token-level Markov Decision Process

To introduce controlled decoding, we first introduce the token-level Markov decision process (MDP) for LLMs. Let $\boldsymbol{x} \in X$ denote the prompt, $\boldsymbol{y} \in Y$ the full response, and $\boldsymbol{y}_{<t} = [y_1, \ldots, y_{t-1}]$ the partial response up to token $t-1$. We define a token-level MDP $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, P, R\}$ in which the state $\boldsymbol{s}_t \in \mathcal{S}$ is the concatenation of the prompt and the generated tokens up to time t, i.e., $\boldsymbol{s}_t = [\boldsymbol{x}, \boldsymbol{y}_{<t}]$. The action space $\mathcal{A}$ corresponds to selecting the next token from the vocabulary $\mathcal{V}$.

Given a state $\boldsymbol{s}_t$, an LLM can be viewed as a policy π that selects the next token by sampling $y_t \sim \pi(\cdot \mid \boldsymbol{s}_t)$. We use ρ to denote the trajectory level distribution induced from the LLM policy π. The state transition is deterministic: once a token is generated, the next state is its concatenation with the current state, i.e., $\boldsymbol{s}_{t+1} = [\boldsymbol{s}_t, y_t]$. The goal of controlled decoding is to align a baseline model ρ_{BL} to a token-level reward function R. As the reward function $r : X \times Y \to \mathbb{R}$ is trained to assign reward to complete responses, the token-level reward is defined as non-zero only when an end of sentence (EOS) token is reached

$$R([\boldsymbol{x}, \boldsymbol{y_{<t}}]) := \begin{cases} 0 & \text{if } y_t \neq \text{EOS}, \\ r([\boldsymbol{x}, \boldsymbol{y_{<t}}]) & \text{if } y_t = \text{EOS}, \end{cases}$$

Given the token-level reward R, we define the value of taking action y_t under a policy π as the expected cumulative reward obtained when future tokens are generated according to π:

$$Q^{\pi}(\boldsymbol{s}_t, y_t) = \mathbb{E}[\sum_i R([\boldsymbol{s}_t, z_i] \mid z_0 = y_t, z_i \sim \pi(\cdot|\boldsymbol{s}_{t+i})] \tag{1}$$

where $s_{t+i} := [\boldsymbol{s}_t, z_0, z_1, \cdots, z_{i-1}]$ and expectation is with respect to the randomness induced by the policy's sampling process. The optimal Q-function from equation 1 is given by

$$Q^*(\boldsymbol{s}_t, y_t) = \max_{\pi} Q^{\pi}(\boldsymbol{s}_t, y_t) \tag{2}$$

In LLM alignment, the goal is not only to maximize the unconstrained objective in (2), but also to keep the policy close to the baseline model that ensures linguistic coherence and fluency.

3.2 Preliminaries on Controlled Decoding

Controlled decoding can be formalized as solving the optimal policy for the following optimization problem:

$$\pi^*_{\text{dec}}(y_t|\boldsymbol{s}_t) := \operatorname*{argmax}_{\pi \in \Pi} E_{z \sim \pi(\cdot|\boldsymbol{s}_t)}[Q^*(\boldsymbol{s}_t, z)] - \beta \mathbb{D}_{\text{KL}}[\pi(\cdot|\boldsymbol{s}_t)||\pi_{\text{BL}}(\cdot|\boldsymbol{s}_t)] \tag{3}$$

where $Q^*(\boldsymbol{s}_t, z)$ is the optimal state–action value from (2). The KL term, weighted by the hyperparameter β, restricts the learned policy to remain close to the baseline model π_{BL}. The above policy has the following closed form solution.

Proposition 1. (3) *has closed-form solution*

$$\pi^*_{dec}(y_t|\boldsymbol{s}_t) = \pi_{\text{BL}}(y_t|\boldsymbol{s}_t) \cdot \frac{\exp\left(\frac{1}{\beta} Q^*(\boldsymbol{s}_t, y_t)\right)}{C_\beta} \tag{4}$$

where $C_\beta = \sum_z \pi_{\text{BL}}(z|s_t) \exp\left(\frac{1}{\beta} Q^*(s_t, z)\right)$ *is the normalizing partion function.*

The major challenge of implementing according to 4 is we don't have oracle access to $Q^*(\boldsymbol{s}_t, y_t):$ it is defined with respect to the optimal policy $\pi^*_{\text{dec}}(y_t|\boldsymbol{s}_t)$, which we cannot sample from. Prior work therefore relies on various approximations. [14] replace $Q^*(\boldsymbol{s}_t, y_t)$ with $r(\boldsymbol{s}_t, y_t)$. [19] approximates $Q^*(\boldsymbol{s}_t, y_t)$ by training a value function that estimates $E_{\tau \sim \rho_{\text{BL}}}[r([\boldsymbol{s}_t, z], \tau)]$, and [5] proposes an importance sampling method to estimate $E_{\tau \sim \rho_r}[r([\boldsymbol{s}_t, z], \tau)]$, where ρ_r is the trajectory distribution induced by an LLM trained with reward function r.

All of these approaches overlook the user's strategic incentives and implicitly assume truthful reward reporting—an assumption that, as we show, warrants careful analysis. Because our principal–agent framework applies broadly across controlled decoding methods, we adopt a trajectory-level formulation as an abstract model of LLM behavior. Given a prompt $\boldsymbol{x}$ and reward function $r(\boldsymbol{x}, \boldsymbol{y})$, the LLM solves the below problem

$$\rho_r(\cdot \mid x) \;=\; \operatorname*{argmax}_{\rho} \mathbb{E}_{\boldsymbol{y} \sim \rho(\cdot|x)}[r(\boldsymbol{x}, \boldsymbol{y})] \;-\; \beta \cdot D_{\text{KL}}\big(\rho(\cdot|\boldsymbol{x}) \,\|\, \rho_{\text{BL}}(\cdot|\boldsymbol{x})\big). \tag{5}$$

The optimization problem (5) has a closed-form solution

$$\rho_r(\boldsymbol{y} \mid \boldsymbol{x}) \;=\; \rho_{\text{BL}}(\boldsymbol{y} \mid \boldsymbol{x}) \cdot \exp\left(\tfrac{1}{\beta} r(\boldsymbol{x}, \boldsymbol{y})\right) \cdot \frac{1}{Z_r(\boldsymbol{x})} \tag{6}$$

, where $Z_r(\boldsymbol{x})$ is again a partition function.

3.3 A game-theoretic critique of controlled decoding

We reinterpret controlled decoding through the lens of a principal–agent model.

- Principal: The entity specifying the reward function—either a platform seeking to align a language model or a platform incorporating user-provided rewards for personalized generation. The principal's true reward function is $r_P(\boldsymbol{x}, \boldsymbol{y})$, while the principal can choose to submit any reward function $r(x, y) \in \mathcal{R}$ to the controlled decoding framework.

- Agent: The language model, which, after receiving the reward function $r(\boldsymbol{x}, \boldsymbol{y})$, generates tokens according to ρ_r specified in (6).

Since the principal selects the reward function first and the LLM subsequently best-responds, controlled decoding can be viewed as a Stackelberg game. The goal of controlled decoding is to maximize $E_{\boldsymbol{y}\sim\rho_r}[r(x, y)]$, the expected reward under the aligned policy ρ_r, while ensuring that the model continues to generate logically coherent responses. **Within the principal–agent formulation, the central question is which reward function the principal should specify to maximize this objective.**

In game theory and mechanism design, when algorithms rely on inputs from strategic humans, humans may have incentives to misreport their information. For instance, an individual might misrepresent their financial profile to increase the likelihood of receiving a loan. A fundamental design objective for algorithms that interact with human users is incentive compatibility, which guarantees that users cannot improve their outcomes by misreporting their information—a property shown to be crucial for the functioning of real-world human-centered systems. In the following example, we show that the standard controlled decoding framework violates this property: when the principal truthfully reports her preferences via reward function, the resulting alignment outcome is suboptimal.

Proposition 2. *Controlled decoding is not incentive compatible.*

Motivating Example: political neutrality with a biased baseline. Given a political prompt $\boldsymbol{x}$, suppose the base model ρ_{BL} only has two possible responses, a left leaning y_1 and a neutral response y_2, with $\rho_{\mathrm{BL}}(y_1|\boldsymbol{x}) = 0.5$ and $\rho_{\mathrm{BL}}(y_2|\boldsymbol{x}) = 0.5$. Consider a principal who favors the neutral response with the following utility profile: $r_P(\boldsymbol{x}, y_1) = 1$ and $r_P(\boldsymbol{x}, y_2) = 2$. The principal can choose to submit a reward function r to aims to maximize $E_{\boldsymbol{y}\sim\rho_r}[r(x, y)]$. Compared to submitting r_P, the principal is better off if she submits $r(\boldsymbol{x}, y_1) = 0$ and $r(\boldsymbol{x}, y_2) = k > 2$, which according to (6) will increase the chance of the model generating the preferred neutral answer [1] When $k \to \infty$, her utility is maximized as the model outputs the neutral response with probability 1. This example leads to the following remarks:

- The example shows the core intuition behind misreporting: the principal can inflate the reward of preferred responses to increase their likelihood, and deflate the reward of undesired ones to suppress them.

- Because the policy is constrained to remain close to the baseline, the decoder may continue to produce left-leaning responses even when the user truthfully reports a reward favoring neutrality. In this way, the KL-regularization term becomes the source of the misalignment, as it encourages adherence to the baseline policy even when its tendencies conflict with the principal's true preferences.

Prior work has documented numerous systematic biases in large language models, so conflicts between the baseline model's tendencies and the principal's utility should be expected in practical deployments. Specifically, existing literature has demonstrated stereotypes in race [1], gender [16, 4], and culture [18], both explicitly and implicitly [2, 17]. These tensions are especially salient in preference dimensions lacking clear normative ground truth—such as individualism versus collectivism—where the base model's generative inclination may diverge substantially from the principal's reward specification.

While truthful reporting may yield suboptimal alignment, naively manipulating the reward function can be even more detrimental. For instance, previous studies show that simply scaling reward can push the decoding policy too far from the base model, often disrupting linguistic coherence and leading to degraded outputs. Therefore, under the principal–agent formulation, our goal is to derive the optimal reward function in a principled manner.

[1] As in prior work on controlled decoding, our framework begins with a given reward function and abstracts away the process by which this reward function is derived from user preferences. Any strategic manipulation by the user can therefore occur either through directly altering the reported reward values or through strategically adjusting their responses in the underlying binary preference data.

3.4 Principal Agent Decoding Formulation

Drawing inspiration from [3], we propose the following reward shaping scheme.

$$\max_{r} \quad \mathbb{E}_{\boldsymbol{y}\sim\rho_r}\big[\, r_P([\boldsymbol{x},\boldsymbol{y}])\,\big] \tag{7}$$

$$\text{s.t.} \quad \rho_r(\cdot\,|\,x) \;=\; \operatorname*{argmax}_{\rho} \mathbb{E}_{\boldsymbol{y}\sim\rho(\cdot|x)}[r(\boldsymbol{x},\boldsymbol{y})] \;-\; \beta \cdot D_{\mathrm{KL}}\big(\rho(\cdot|\boldsymbol{x}) \,\|\, \rho_{\mathrm{BL}}(\cdot|\boldsymbol{x})\big). \tag{C1}$$

$$0 \le r(\boldsymbol{x},\boldsymbol{y}) \;\le\; B.\forall \boldsymbol{y} \tag{C2}$$

The principal's objective is to maximize their expected utility, subject to two constraints grounded in alignment application [2]:

- Constraint C1 is the resulting model's controlled decoding policy after receiving the reward function $r(\boldsymbol{x},\boldsymbol{y})$.
- Constraint C2 follows the recommendation of [7], which emphasizes the importance of bounding the reward function to reduce the risk of reward hacking in LLM alignment. The upper bound B is a hyperparameter and can be viewed as a "budget constraint" that limits the extent to which the principal can pull the aligned policy away from the baseline model.

The optimal solution to (7) admits the following threshold structure.

Theorem 3. *Let's define*

$$r_m(x,y) := \begin{cases} B & \text{if } \; r_P(x,y) \ge m(x) \\ 0 & \text{if } \; r_P(x,y) < m(x). \end{cases}$$

and $g_{\boldsymbol{x}}(m) = \mathbb{E}_{\boldsymbol{y}'\sim\rho_{r_m}(\cdot|\boldsymbol{x})}[r_P(x,y')]$. *Let* $m^*(x)$ *be a solution to* $g_x(m) = m$, *for any prompt* x. *With minor assumptions, the reward function* r_{m^*} *defined by* $m^*(x)$ *is an optimal reward function for the principal.*

Proof. We defer the proof to Appendix 7.1. □

The solution indicates if the principal values a response $\boldsymbol{y}$ above a certain **prompt-dependent threshold**, its reward should be set to the upper bound B; otherwise, the response should be penalized. We examine the above solution:

- This aligns with the intuition from our motivating example: the principal should increase the reward for preferred responses and apply penalties to those that are less preferred. The resulting reward function exhibits the geometric pattern of a rapid rise followed by convergence, a pattern that is shown to be desirable for alignment [7]. [3] However, compared to prior work, the principal–agent perspective allows us to select a principled and informative threshold rather than relying on an ad hoc statistic such as the mean.
- The entropy regularization in the agent's best response is crucial for obtaining the closed-form solution in (7), a feature that is essential for alignment applications. In contrast, without entropy regularization, solving principal–agent reward shaping in general MDPs is NP-hard, and even in structured MDPs the computational cost grows with the number of possible trajectories (responses) [3]. In our setting, if the maximum response length is l (128 in our experiments), the induced trajetcory space has size $|\mathcal{V}|^l$, which is prohibitively large for any practical computation.
- The **prompt dependent threshold** requires sampling access to $\rho_{r_{m^*}}$, yet obtaining $\rho_{r_{m^*}}$ itself presupposes knowledge of r_{m^*}. This circular dependence creates a fixed point challenge, which we address in the next section by developing algorithms that approximate the threshold.

When there is no confusion, we write r^* for $r_{m^*}(\boldsymbol{x},\boldsymbol{y})$. After obtaining r^*, we use the modified reward function in the controlled decoding algorithm, and we term our framework **Principal Agent Decoding (PAD)**. Before introducing the algorithm that computes the threshold, we first analyze the theoretical properties of the optimal reward $r^*(\boldsymbol{x},\boldsymbol{y})$.

[2] This formulation may be of independent interest, as it extends principal–agent models to settings where the agent operates under a KL-based entropy constraint—a structure that arises not only in LLM alignment but also in models of bounded rationality.

[3] Unlike [7] which studies reward shaping in RLHF, we do not require the reward function to be smooth, as controlled decoding never relies on computing its gradients.

Proposition 4. *Under the PAD algorithm, the principal's best strategy is to report their true reward function $r_P(\boldsymbol{x}, \boldsymbol{y})$; that is, PAD is incentive compatible.*

Proof. The proof is deferred to the Appendix. □

Conjecture 5. Assume reward satisfies $0 \leq r \leq r_{\max}$ and response length is bounded by l, then the divergence to baseline policy is given by

$$\mathbb{D}_{KL}(\rho_{\mathrm{BL}}(\cdot|\boldsymbol{x}), \rho_r(\cdot|\boldsymbol{x})) \leq C(\beta, l) r_{\max}$$

, where $C(\beta, l)$ is a constant only depends on the hyperparameter β and response length l.

Proof. We are pretty sure this can be proved, probably sufficient to drawing techniques directly from [5], but we haven't had the chance to formalize the result yet. We should be able to obtain the result before the poster deadline. □

Suppose Conjecture 5 is true, it immediately implies the following corollary.

Corollary 6. *Since r^* is bounded above by B, we have*

$$\mathbb{D}_{KL}(\rho_{\mathrm{BL}}(\cdot|\boldsymbol{x}), \rho_{r^*}(\cdot|\boldsymbol{x})) \leq C(\beta, l) B$$

This corollary, if correct, implies that the resulting decoding policy remains close to the baseline model and therefore preserves its ability to produce coherent responses.

4 Computing the optimal reward function

We now describe how to compute the prompt-dependent threshold, defined as the fixed point of the equation $g_{\boldsymbol{x}}(m) = m$, where where $m(\boldsymbol{x})$ denotes this threshold for prompt $\boldsymbol{x}$. For notational convenience, let $r_i = r(\boldsymbol{x}, \boldsymbol{y}_i)$. Suppose for a given prompt $\boldsymbol{x}$, there are $N \leq \mathcal{V}^l$ possible responses for it under policy ρ_{BL}. We call them $\boldsymbol{y}_1, \cdots, \boldsymbol{y}_N$ whose index are sorted according to reward, i.e., $r_1 < r_2 \cdots < r_N$.

Lemma 7. *We define $w_-(m) := \rho_{\rho_{\mathrm{BL}}}(\boldsymbol{y}_i|\boldsymbol{x})$ if $r_i < m$, and $w_+(m) := c\rho_{\mathrm{BL}}(y_i|x)$ if $r_i \geq m$, with $c = \exp\left(\frac{B}{\beta}\right) > 1$. Expand the expectation in $g(m)$'s definition, we have:*

$$g_{\boldsymbol{x}}(m) = \frac{\sum_{y_i : r_P(x, y_i) \geq m} w_+(x, y_i) r_P(x, y_i) + \sum_{y_i : r_P(x, y_i) < m} w_-(x, y_i) r_P(x, y_i)}{\sum_{y_i : r_P(x, y_i) \geq m} w_+(x, y_i) + \sum_{y_i : r_P(x, y_i) < m} w_-(x, y_i)}$$

Definition 8. Let us define the helper function

$$F(m) = \sum_{i=1}^{N} w_i(m) \cdot (r_i - m).$$

where $w_i(m) = w_-(m)$ if $r_i < m$, and $w_i(m) = w_+(m)$ if $r_i \geq m$.

Theorem 9. *Consider the function $H(m) = g_{\boldsymbol{x}}(m) - m$, we show that $H(m)$ has a unique root on the interval $m \in [r_1, r_N]$.*

Proof. The proof proceeds in two parts: First, we show that $H(m) = 0$ iff $F(m) = 0$. Then, we show that $F(m)$ is monotonically decreasing on the interval $m \in [r_1, r_N]$ and has a unique root. The full proof if deferred to the Appendix 7.3. □

Hence, to find the cutoff, one only need to run a root finding algorithm to compute the root for helper function $F(m)$. In fact, one can do bisection to exploit the monotonicity of F. Another challenge we have is the space of possible response trajectory is very large, and in practice we cannot sample all possible trajectory from ρ_{BL}. To address this problem, we propose the following Monte Carlo estimator.

Algorithm 1 Principal–Agent Decoding (PAD) Training Procedure

Require: base model ρ_{BL}, principal reward r_P, reward upper bound B, entropy regularization coefficient β, sample size $M = 10$, offline dataset $\mathcal{D}$ with prompts

Ensure: Value function for PAD-controlled decoding

1: **Step 1: Generate offline dataset and perform reward shaping**

2: For each given prompt $\boldsymbol{x} \sim D$, sample $\{\boldsymbol{y}_i\}_{i=1}^M$ from $\rho_{\text{BL}}(\cdot|x)$ and their corresponding reward $\{r_i\}_{i=1}^M$ under r_P, construct the Monte Carlo estimator for the helper function 8 and run bisection to obtain its root, i.e., estimated cutoff $\hat{m}(\boldsymbol{x})$

Perform the following reward shaping:

$$r^*(\boldsymbol{x}, \boldsymbol{y}_i) \;=\; \begin{cases} B, & \text{if } r_i \geq \hat{m}(x), \\ 0, & \text{if } r_i < \hat{m}(x). \end{cases}$$

3: **Step 2: Train the value network using Bellman equation** With $(x, \{y_i\}_{i=1}^M)$ and the modified reward $\{r_i^*\}_{i=1}^M$, the problem is reduced to the same problem as standard controlled decoding, we train the action value function the same way as in [19]

4: **return** Q_{PAD}

Monte Carlo Estimator We sample $\{\boldsymbol{y}_i\}_{i=1}^M$ from $\rho_{\text{BL}}(\cdot|x)$. Then

$$\hat{F}_M(m) = \frac{1}{M} \sum_{i=1}^{M} (r(x, y_i) - m)\big(1 + (c-1)\,\mathbf{1}\{r(x, y_i) > m\}\big) \tag{8}$$

with $c = \exp\!\left(\frac{B}{\beta}\right)$. Therefore, we find the approximate cutoff by finding the root for the above Monte Carlo estimator. We formally introduce our algorithm Principal Agent Decoding (PAD) in Algorithm 1.

5 Experiment Results

In this section, we present empirical results evaluating the effectiveness of our approach. We focus on aligning large language models (LLMs) for helpfulness and reducing harmful behavior—two central objectives in LLM alignment.

5.1 Experiment Setup

We evaluate our method on Dahoas HH-RLHF dataset [4], which is popular for alignment task. Each sample in the dataset contains a prompt and two responses, one is chosen and other is rejected. For the base model, we adopt LLama3-8b-instruct fine tuned [5], and for the reward model r_P, we use a public reward model UltraRM 13B for ease of replication [6]. For controlled decoding, the hyperparameter is β, and for PAD, the hyperparameter is β and the reward upper bound B. We select the hyperparameter on a held-out validation set of 622 prompts, and then asks the decoding policy to answer the first 622 prompts in the test set and score the responses. Following prior literature in inference time alignment [14, 5], we report average reward, coherence, diversity, and win-tie rate. We introduce the metrics used:

- Diversity: This metric assesses a model's ability to produce linguistically diverse text by measuring the frequency of repeated n-grams.
- Coherence: We quantify semantic consistency between each prompt and its generated response by computing the cosine similarity of their SimCSE embeddings [22].
- Average reward: Mean reward of the generation, scored by r_P.

[4] https://huggingface.co/datasets/Dahoas/full-hh-rlhf
[5] https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct
[6] https://huggingface.co/openbmb/UltraRM-13b

Dataset	Backbone	Model	Diversity ↑	Coherence ↑	Avg Reward ↑	Win-Tie Rate (%) ↑
HH-RLHF	Llama3-8B	Base	0.808	0.589	-4.97	78
		CD (prefix)	0.796	0.607	-4.89	83
		PAD	0.787	0.593	-4.73	84

Table 1: Performance comparison on HH-RLHF using the Llama3-8B-Instruct as the base model. Win rate is evaluated by GPT-4 as the rate at which the model's response is rated better than or achieve ties with the preferred response in the dataset.

- GPT win-tie rate: For 100 randomly selected prompt-responses generated by each method, we give the response to GPT-4 as a judge and ask it to compare it against the preferred answer in the original HH-RLHF dataset. We randomly shuffle the order in which we present the response to GPT-4 judge to prevent positional bias. We report the percentage which each method either achieves a win or a tie against the original method.

5.2 Experiment Result Analysis

We present the result in Table 1. Our method achieves roughly twice the reward gain obtained by controlled decoding while preserving output coherence. Although diversity decreases slightly, this pattern aligns with prior findings showing that inference-time alignment methods can sometimes reduce diversity. The improvement in win-tie rate is modest, which is a bit disappointing but understandable given that the baseline model is already highly aligned—outperforming the preferred human response in 78% of cases in the HH-RLHF dataset.

We list some possible extensions and improvements to our experiment section:

- The primary motivation for our framework is to address situations in which the model's intrinsic preferences conflict with those encoded in the human reward model. However, in this experiment the base model is already highly aligned (as shown by the 78% win-tie rate). Consequently, meaningful evaluation requires alignment tasks that are challenging for the base model or datasets in which such preference conflicts naturally arise—for example, cases where the baseline model favors collectivist statements while the principal prefers individualistic ones.
- We currently take 10 samples for each prompt when we construct the offline dataset for Q function training. Maybe more sample is needed for giving a good estimation of the threshold.
- There is one inconsistency in the implementation between the experiment and the theory, as we adopts greedy decoding rather than sampling. We will change it so that the theory and experiments are fully aligned.
- We need to test our method on more base model and reward model combinations and perform ablation studies.
- It's worth trying a softer version of the Principal Agent Decoding, where the reward for y_i is shaped to $B\sigma(\alpha(r(\boldsymbol{x}, \boldsymbol{y}_i) - m(\boldsymbol{x})))$, where α is a hyperparameter. When $\alpha \to 0$, this converges to no alignment effect, and when $\alpha \to \infty$, this converges to the hard shaping, so this soft version effectively interpolates between no alignment and hard shaping. The sigmoid form is shown to be optimal in [7] for all smooth reward functions, and the difference here is our principal-agent formulation helps to select an informative threshold rather than ad hoc statistics such as mean of reward.

5.3 Reward Dominance across β and hyperparameter study

Table 1 reports results for each method using its optimal hyperparameters. However, our reward-shaping scheme is derived under the assumption of a fixed value of β in the language-model best-response step. This implies a stronger statement: for any given β, our shaped reward should yield higher performance than using the unshaped reward. This behavior is confirmed in Figure 1, where we vary β (plotted as $1/\beta$ on the x-axis) and report the corresponding average reward. Note when $\frac{1}{\beta}$ is large—meaning the KL penalty is weak—controlled decoding tends to oversteer, leading to degraded performance. In contrast, PAD avoids this issue because the shaped reward is explicitly bounded. In this regime, smaller values of the reward cap B yield better

β	5.0	2.0	1.43	1.0	0.83	0.67	0.5	0.4	0.33	0.25	0.20	0.17
optimal B	20	20	20	20	12	5	10	5	12	10	10	10

Table 2: Mapping between $1/\lambda$ and selected hard reward bounds B.

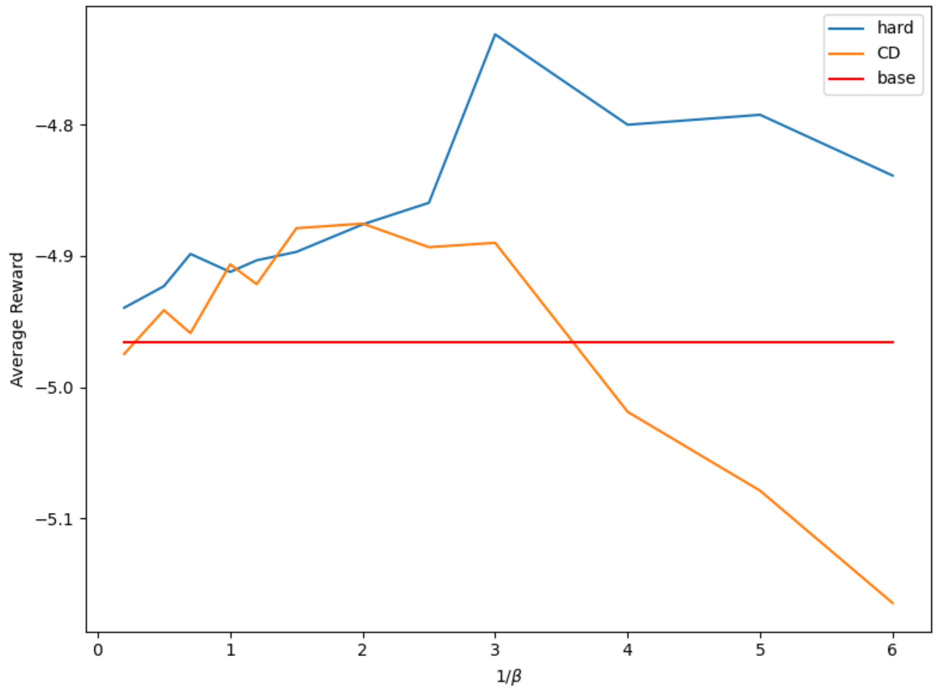

Figure 1: The average reward of each method vs different hyperaprameter $\frac{1}{\beta}$.

performance. We report the optimal hyperparameter B for each corresponding β in Table 2, and it confirms our intuition, when β is small, a tighter reward bound B is preferred.

6 Safety Applications of Inference-time Alignment

While established approaches like supervised fine-tuning and RLHF remain the current primary methods for aligning large language models, inference-time techniques offer distinct advantages in scenarios requiring rapid adaptation without retraining overhead. The Principal-Agent Decoding framework addresses a gap in current alignment methods: the ability to specify and maintain precise, personalized alignment objectives at deployment time. By solving the incentive compatibility problem through contract design, our method enables portable and mathematically grounded alignment that prevents reward hacking while remaining faithful to specified preferences.

This capability enables applications where alignment must be both flexible and trustworthy. For instance, in educational settings, teachers could design specific pedagogical reward profiles, balancing encouragement with correction, that adapt AI tutors to individual learning styles without generic model behavior. For personal AI assistants acting as decision proxies in decentralized governance systems, this approach could ensure outputs remain aligned with users' evolving preferences without requiring constant retraining, making alignment both lightweight and privacy-preserving. In multi-stakeholder scenarios such as AI mediators or content moderation, the framework enables explicit encoding of competing objectives with formal guarantees that models won't arbitrarily prioritize one party over another.

More broadly, Principal-Agent Decoding provides a governance layer for scenarios where AI agents must reliably represent individual preferences rather than regress to population averages, whether in personalized healthcare, legal negotiation, or collaborative human-robot systems. As AI systems increasingly serve as extensions of individual agency, frameworks that guarantee alignment to specific user values become essential safety infrastructure.

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7 Appendix

7.1 Proof of Theorem 3

We first prove the following lemma.

Lemma 10.
$$\frac{\partial \rho_r(y'|x)}{\partial r(x,y)} = \frac{1}{\beta}\rho_r(y'|x) \cdot (\delta_{y',y} - \rho_r(y|x)).$$

Proof. **Case 1:** $y' = y$**.** We know that

$$\rho_r(y' \mid x) \;=\; \frac{\rho_{\mathrm{BL}}(y' \mid x)\, \exp\Bigl(\frac{1}{\beta}\left(r(x,y') - r_{\mathrm{BL}}(x,y')\right)\Bigr)}{\tilde{Z}_r(x)},$$

where $\tilde{Z}_r(x)$ is the partition function. Let's first work on the partial derivative of the numerator:

$$\begin{aligned}
&\frac{\partial \rho_{\mathrm{BL}}(y' \mid x)\, \exp\Bigl(\frac{1}{\beta}\left(r(x,y') - r_{\mathrm{BL}}(x,y')\right)\Bigr)}{\partial r(x,y)} \\
&= \frac{\rho_{\mathrm{BL}}(y' \mid x)}{\exp\Bigl(\frac{1}{\beta} r_{\mathrm{BL}}(x,y')\Bigr)} \cdot \frac{1}{\beta}\exp\left(\frac{1}{\beta} r(x,y')\right) \\
&= \frac{1}{\beta} \cdot \rho_{\mathrm{BL}}(y' \mid x)\, \exp\Bigl(\frac{1}{\beta}\left(r(x,y') - r_{\mathrm{BL}}(x,y')\right)\Bigr)
\end{aligned}$$

Then we work on the partial derivative of the denominator, with $\tilde{Z}_r(x) = \sum_{\tilde{y}} \rho_{\mathrm{BL}}(\tilde{y} \mid x)\, \exp\Bigl(\frac{1}{\beta}\left(r(x,\tilde{y}) - r_{\mathrm{BL}}(x,\tilde{y})\right)\Bigr)$.

$$\begin{aligned}
&\frac{\partial \tilde{Z}_r(x)}{\partial r(x,y)} \\
&= \frac{1}{\beta} \cdot \rho_{\mathrm{BL}}(y \mid x)\, \exp\Bigl(\frac{1}{\beta}\left(r(x,y) - r_{\mathrm{BL}}(x,y)\right)\Bigr) \\
&= \frac{1}{\beta}\tilde{Z}_r(x)\frac{\rho_{\mathrm{BL}}(y \mid x)\, \exp\Bigl(\frac{1}{\beta}\left(r(x,y) - r_{\mathrm{BL}}(x,y)\right)\Bigr)}{\tilde{Z}_r(x)} \\
&= \frac{1}{\beta}\tilde{Z}_r(x)\rho_r(y|x)
\end{aligned}$$

Let $w_{x,y'} = \rho_{\mathrm{BL}}(y' \mid x)\, \exp\Bigl(\frac{1}{\beta}\left(r(x,y') - r_{\mathrm{BL}}(x,y')\right)\Bigr)$. Then by quotient rule, we would have:

$$\begin{aligned}
\frac{\partial \rho_r(y'|x)}{\partial r(x,y)} &= \frac{\tilde{Z}_r(x) \cdot \frac{1}{\beta} \cdot w_{x,y'} - w_{x,y'} \cdot \rho_r(y|x) \cdot \frac{1}{\beta} \cdot \tilde{Z}_r(x)}{\tilde{Z}_r(x)^2} \\
&= \frac{1}{\beta}\rho_r(y'|x) \cdot (1 - \rho_r(y|x))
\end{aligned}$$

Case 2: $y' \neq y$**.** Follow some similar algebra as above, we obtain the following solution:

$$\frac{\partial \rho_r(y'|x)}{\partial r(x,\tau)} = \frac{1}{\beta}\rho_r(y'|x) \cdot -\rho_r(y|x)$$

Combined the two results together, we have:

$$\frac{\partial \rho_r(y'|x)}{\partial r(x,y)} = \frac{1}{\beta}\rho_r(y'|x) \cdot (\delta_{y',y} - \rho_r(y|x))$$

□

Theorem 11. *For a given prompt x and a specific answer trajectory y_i, the solution to Program* **??** *is*

$$r(x,y_i)=\begin{cases}B & \text{if } \ r_P(x,y_i)\geq \mathbb{E}_{y'\sim\rho_r}r_P(x,y')\\ 0 & \text{if } \ r_P(x,y_i)< \mathbb{E}_{y'\sim\rho_r}r_P(x,y')\end{cases}$$

Proof. Let $J(r)=\mathbb{E}_{y\sim\rho_r(\cdot|x)}\big[\,r_P(x,y)\,\big]$. Denote $r_i:=r(x,y_i)$. Let there be N possible response trajectory in total. Set the Lagrangian: $L(r,\mu,\nu)=J(r)+\sum_{i=1}^N\mu_i r_i+\sum_{i=1}^N\nu_i(B-r_i)$. Stationarity gives

$$\frac{\partial J}{\partial r_i}+\mu_i-\nu_i=0 \tag{9}$$

where by Lemma 10, we have

$$\begin{aligned}\frac{\partial J}{\partial r_i}&=\sum_{y'}r_P(x,y')\cdot\frac{\partial\rho_r(y'|x)}{\partial r_i}\\ &=\sum_{y'}r_P(x,y')\cdot\frac{1}{\beta}\cdot\rho_r(y'|x)\cdot(\delta_{y',y}-\rho_r(y|x))\\ &=\frac{1}{\beta}[r_P(x,y_i)\rho_r(y_i|x)-\rho_r(y_i|x)\sum_{y'}r_P(x,y')\rho_r(y'|x)]\\ &=\frac{1}{\beta}\rho_r(y_i|x)\Big(r_P(x,y_i)-\mathbb{E}_{y'\sim\rho_r}[r_P(x,y')]\Big).\end{aligned}$$

Complementary slackness and feasibility read

$$\begin{cases}\mu_i\cdot r_i=0, & \nu_i(B-r_i)=0,\\ \mu_i\geq 0, & \nu_i\geq 0,\\ 0\leq r_i\leq B\end{cases}$$

Interior Solution: Let's consider the interior solution $0<r(x,y_i)<B$. This means both $\mu_i=\nu_i=0$. By the stationary condition, we have

$$r_P(x,y_i)-\mathbb{E}_{y'\sim\rho_r}[r_P(x,y')]=0$$

Since r_P is not a constant, unless $r_P(x,y_i)=\mathbb{E}_{y'\sim\rho_r(\cdot|x)}[r_P(x,y')]$, we won't have an interior solution.

Lower Bound $r(x,y_i)=0$: By complementary slackness, this implies that $\mu_i\geq 0$ and $\nu_i=0$. Because $B-r(x,y_i)>0$, it must be the case that $\nu_i=0$. Plug in all these information into the stationarity condition, we have:

$$\frac{1}{\beta}\rho_r(y_i|x)\Big(r_P(x,y_i)-\mathbb{E}_{y'\sim\rho_r(\cdot|x)}[r_P(x,y')]\Big)+\mu_i=0,$$

which implies

$$\frac{1}{\beta}\rho_r(y_i|x)\Big(r_P(x,y_i)-\mathbb{E}_{y'\sim\rho_r(\cdot|x)}[r_P(x,y')]\Big)=-\mu_i\leq 0$$

Hence $r_P(x,y_i)\leq\mathbb{E}_{y'\sim\rho_r(\cdot|x)}[r_P(x,y')]$.

Upper Bound $r(x,\tau_i)=B$: By complementary slackness, this implies that $\mu_i=0$ and $\nu_i\geq 0$. Because $r(x,y_i)=B$, it must be the case that $\mu_i=0$. Plug in all these information into stationary condition, we have:

$$\frac{1}{\beta}\rho_r(y_i|x)\Big(r_P(x,y_i)-\mathbb{E}_{y'\sim\rho_r(\cdot|x)}[r_P(x,y')]\Big)-\nu_i=0,$$

which implies

$$\frac{1}{\beta}\rho_r(y_i|x)\Big(r_P(x,y_i)-\mathbb{E}_{y'\sim\rho_r(\cdot|x)}[r_P(x,y')]\Big)=\nu_i\geq 0.$$

Hence $r_P(x,y_i)\geq\mathbb{E}_{y'\sim\rho_r(\cdot|x)}[r_P(x,y')]$. Note that if $r_P(x,y_i)>\mathbb{E}_{y'\sim\rho_r(\cdot|x)}[r_P(x,y')]$, it can only correspond to the case $r(x,y_i)=B$; and if $r_P(x,y_i)<\mathbb{E}_{y'\sim\rho_r(\cdot|x)}[r_P(x,y')]$, it can only correspond to the case $r(x,y_i)=0$. In summary,

$$r(x,y_i)=\begin{cases}B & \text{if } \ r_P(x,y_i)\geq \mathbb{E}_{y'\sim\rho_r}r_P(x,y')\\ 0 & \text{if } \ r_P(x,y_i)< \mathbb{E}_{y'\sim\rho_r}r_P(x,y')\end{cases}$$

□

7.2 Proof of Proposition 4

Proof. Let $u(r_P, r_S)$ be the utility of a principal with utility function r_P but report utility function r_S to the controlled decoding framework. Hence, we want to show that $u(r_P, r_P) \geq u(r_P, r_S)\ \forall r_S$,

Suppose the user reports utility r_P truthfully, and the solution to Program 7 under truthful report is r^*. Then the user receives utility $u(r_P, r_P) = \max_r E_{y\sim\rho_r} r_P(x, y)$. If the user reports r_S, then the user will receive utility $E_{\tau\sim\rho_{r'}} r_P(\tau) = u(r_P, r_S)$ where r' be the solution of the below program

$$\begin{aligned}
&\max_{r} && \mathbb{E}_{\tau\sim\rho_r}\big[\, r_S[x,y]\,\big] \\
&\text{s.t.} && \rho_r(y \mid x) \;=\; \rho_{\mathrm{BL}}(y \mid x)\,\exp\!\Big(\tfrac{1}{\beta}\,\big(r(x,y) - r_{\mathrm{BL}}(x,y)\big)\Big)\cdot\frac{Z_{BL}(x)}{Z_r(x)}, \\
& && 0 \leq r(x,y) \;\leq\; B.\forall y
\end{aligned}$$

Note since r^* and r' are solutions to two optimization problems with the same feasibility region (same budget constraint), we have $E_{\tau\sim\rho_{r^*}} r_P(\tau) = \max_r \mathbb{E}_{\tau\sim\rho_r}\big[\, r_P[\tau]\,\big] \geq E_{\tau\sim\rho_{r'}} r_P(\tau)$. This complete the proof. $\square$

7.3 Proof of Theorem 9

Proof. The proof proceeds in two parts: First, we define a helper function $F(m)$ and show that $H(m) = 0$ iff $F(m) = 0$. Then, we show that $F(m)$ is monotonically decreasing and has a unique root.

To prove the first part, note that

$$\begin{aligned}
F(m) &= \sum_{i=1}^{N} w_i(m)\cdot(r_i - m) \\
&= \sum_{i=1}^{N} w_i(m)\cdot r_i - \left(\sum_{i=1}^{N} w_i(m)\right)\cdot m \\
&= \left(\sum_{i=1}^{N} w_i(m)\right)\cdot g(m) - \left(\sum_{i=1}^{N} w_i(m)\right)\cdot m \\
&= \left(\sum_{i=1}^{N} w_i(m)\right)\cdot (g(m) - m)
\end{aligned}$$

Because $\sum_{i=1}^{N} w_i(m) > 0$, it must be the case that $F(m) = 0$ if and only if $g(m) - m = 0$.

To prove the second part, we first show that $F(m)$ is continuous and monotonically decreasing in m. We devide into two possible cases.

Case 1 Consider any two adjacent rewards r_j and r_{j+1}, consider the case when $m \in (r_j, r_{j+1})$. It is obvious that any $m \in (r_j, r_{j+1})$ corresponds to the same cutoff, so $g(m)$ is a constant and $g(m) - m$ is monotonically decreasing. $\sum_{i=1}^{N} w_i(m)$ is the same constant given the cutoff remains the same, so $F(m)$ is continuous and decreasing between any two adjacent rewards.

Case 2 Consider $m = r_k$ for any $k \in [2, \cdots, N-1]$. Note that

$$F(m) = \sum_{i\leq k-1} w_i(r_i - m) + \sum_{i\geq k+1} w_i(r_i - m)$$

When $t \to r_k^-$, we have

$$\begin{aligned}
\lim_{t\to r_k^-} F(t) &= \lim_{t\to r_k^-} \sum_{i\leq k-1} w_i(r_i - t) + \rho_{\mathrm{BL}}(y_k|x)(r_k - t) + \sum_{i\geq k+1} w_i(r_i - t) \\
&= F(r_k)
\end{aligned}$$

where the second equality is obtained because $r_k - t \to 0$ when $t \to r_k^-$.

When $t \to r_k^+$, we have

$$\begin{aligned}\lim_{t \to r_k^+} F(t) &= \lim_{t \to r_k^+} \sum_{i \le k-1} w_i(r_i - t) + c\rho_{\mathrm{BL}}(y_k|x)(r_k - t) + \sum_{i \ge k+1} w_i(r_i - t) \\ &= F(r_k)\end{aligned}$$

where the second equality is obtained because $r_k - t \to 0$ when $t \to r_k^+$. Hence $\lim_{t \to r_k^-} F(t) = F(r_k) = \lim_{t \to r_k^+} F(t)$, which implies that F is also continuous at all the intermediate reward.

Combine both cases, $F(m)$ is continuous and monotonically decreasing on $[r_1, r_N]$. Since rewards are sorted, we can easily see that

$$F(r_1) = \sum_{i=1}^{N} w_i(m)(r_i - r_1) > 0$$

and

$$F(r_N) = \sum_{i=1}^{N} w_i(m)(r_i - r_N) < 0$$

Since F is continuous and two ends have opposite signs, by intermediate value theorem, $F(m)$ must have at least one root. By monotonicity of F, the root is unique. This completes the proof. $\square$